1 Business Understanding

Objective

The goal of this project is to analyze Formula 1 (F1) data to gain insights into race performance, constructor strategies, and driver effectiveness. The analysis aims to provide valuable insights to stakeholders, such as team managers, sponsors, and sports analysts, to help them make informed decisions about future races, sponsorship opportunities, and driver selections.

Key Stakeholders

- **Team Managers**: Interested in understanding the performance trends of their drivers and constructors to strategize for upcoming races.
- **Sponsors**: Looking for data-driven insights to identify the most promising teams and drivers for sponsorship.
- **Sports Analysts**: Seeking in-depth analysis of race outcomes, driver performance, and constructor strategies to provide expert commentary and predictions.
- **F1 Enthusiasts**: Fans who are keen on understanding the historical performance of their favorite drivers and teams.

2 Data Understanding

Overview

The dataset contains four key components:

Results Dataset

- **Description**: Contains detailed information about race results, including positions, points, and other race-specific details.
- **Key Columns**: `race_id`, `driver_id`, `constructor_id`, `position`, `points`, `laps`, `time`, `status`.

Constructors Dataset

- **Description**: Provides details about the constructors (teams) involved in F1 races.
- **Key Columns**: `constructor_id`, `constructor_name`, `nationality`.

Race Dataset

- **Description**: Includes information about the races themselves, such as location, date, and circuit details.
- **Key Columns**: `race_id`, `year`, `round`, `circuit_id`, `date`, `time`, `name`.

Drivers Dataset

- **Description**: Contains information about the drivers, including their personal details and career stats.
- **Key Columns**: `driver_id`, `driver_name`, `dob`, `nationality`, `wins`, `poles`.

3 Data preparation

```
# import the necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

3.1 Reading the Data

```
# reading  the constructors datasets
constructors = pd.read_csv("constructors.csv")
constructors.head()
```

	constructorId	constructorRef	name	nationality	url
0	1	mclaren	McLaren	British	http://en.wikipedia.org/wiki/McLaren
1	2	bmw_sauber	BMW Sauber	German	http://en.wikipedia.org/wiki/BMW_Sauber
2	3	williams	Williams	British	http://en.wikipedia.org/wiki/Williams_Grand_Pr...
3	4	renault	Renault	French	http://en.wikipedia.org/wiki/Renault_in_Formul...
4	5	toro_rosso	Toro Rosso	Italian	http://en.wikipedia.org/wiki/Scuderia_Toro_Rosso

```
# reading the result dataset
results = pd.read_csv("results.csv")
results.head()
```

	resultId	raceId	driverId	constructorId	number	grid	position	positionText	positionOrder	points	laps	
0	1	18	1	1	22	1	1	1	1	10.0	58	1:34
1	2	18	2	2	3	5	2	2	2	8.0	58	
2	3	18	3	3	7	7	3	3	3	6.0	58	
3	4	18	4	4	5	11	4	4	4	5.0	58	-
4	5	18	5	1	23	3	5	5	5	4.0	58	-

```
# reading the race dataset
race = pd.read_csv("races.csv")
race.head()
```

	raceId	year	round	circuitId	name	date	time	url	fp1_date	fp1_time
0	1	2009	1	1	Australian Grand Prix	2009-03-29	06:00:00	http://en.wikipedia.org/wiki/2009_Australian_G...	\N	\N
1	2	2009	2	2	Malaysian Grand Prix	2009-04-05	09:00:00	http://en.wikipedia.org/wiki/2009_Malaysian_Gr...	\N	\N
2	3	2009	3	17	Chinese Grand Prix	2009-04-19	07:00:00	http://en.wikipedia.org/wiki/2009_Chinese_Gran...	\N	\N
3	4	2009	4	3	Bahrain Grand Prix	2009-04-26	12:00:00	http://en.wikipedia.org/wiki/2009_Bahrain_Gran...	\N	\N
4	5	2009	5	4	Spanish Grand Prix	2009-05-10	12:00:00	http://en.wikipedia.org/wiki/2009_Spanish_Gran...	\N	\N

```
# reading the drivers dataset
drivers = pd.read_csv("drivers.csv")
drivers.head()
```

	driverId	driverRef	number	code	forename	surname	dob	nationality	url
0	1	hamilton	44	HAM	Lewis	Hamilton	1985-01-07	British	http://en.wikipedia.org/wiki/Lewis_Hamilton
1	2	heidfeld	\N	HEI	Nick	Heidfeld	1977-05-10	German	http://en.wikipedia.org/wiki/Nick_Heidfeld
2	3	rosberg	6	ROS	Nico	Rosberg	1985-06-27	German	http://en.wikipedia.org/wiki/Nico_Rosberg

3.2 **Data Exploration**

We will focus on all the dataset provided for this project first we will have to merge all the dataset together

```
# merging the data sets
df = pd.merge(results,race[["raceId", "year", "name", "round"]],on="raceId",how="left")
df = pd.merge(df,drivers[["driverId","driverRef","nationality"]], on= "driverId",how="left")
df = pd.merge(df,constructors[["constructorId","name","nationality"]],on= "constructorId",how="left")
```

```
df.head()
```

	resultId	raceId	driverId	constructorId	number	grid	position	positionText	positionOrder	points	...	fast
0	1	18	1	1	22	1	1	1	1	10.0	...	
1	2	18	2	2	3	5	2	2	2	8.0	...	
2	3	18	3	3	7	7	3	3	3	6.0	...	
3	4	18	4	4	5	11	4	4	4	5.0	...	
4	5	18	5	1	23	3	5	5	5	4.0	...	

5 rows × 25 columns

```
# summary stats
df.describe()
```

	resultId	raceId	driverId	constructorId	grid	positionOrder	points	laps	
count	26519.000000	26519.000000	26519.000000	26519.000000	26519.000000	26519.000000	26519.000000	26519.000000	265
mean	13260.940986	546.376560	274.357291	49.801161	11.145820	12.814812	1.959578	46.228251	
std	7656.813206	309.642244	279.275606	61.091426	7.213453	7.677869	4.306475	29.577860	
min	1.000000	1.000000	1.000000	1.000000	0.000000	1.000000	0.000000	0.000000	
25%	6630.500000	298.000000	57.000000	6.000000	5.000000	6.000000	0.000000	23.000000	
50%	13260.000000	527.000000	170.000000	25.000000	11.000000	12.000000	0.000000	53.000000	
75%	19889.500000	803.000000	385.000000	60.000000	17.000000	18.000000	2.000000	66.000000	
max	26524.000000	1132.000000	860.000000	215.000000	34.000000	39.000000	50.000000	200.000000	1

Data cleaning

From our data we notice that there are some columns that are irrelevant to our analysis our first step will be to drop them

```
# dropping unnecessary columns from our dataset
df.drop(["resultId","raceId","driverId","constructorId","number","position","positionText","statusId"],axis=1,in
```

The next step will be renaming the columns for clarity,redability and consistency

```
# Renaming the columns for clarity and consistency
df.rename(columns={
    "rank" : "fastestLapRank",
    "name_x": "gp_name",
    "driverRef": "driver",
    "name_y": "constructors_name",
    "nationality_x": "drivers_nationality",
    "nationality_y": "constructors_nationality"
}, inplace=True)
```

```
# sorting the dataset by year round and positionOrder
df = df.sort_values(by=['year',"round","positionOrder"],ascending=[False,True,True])
df.head()
```

	grid	positionOrder	points	laps	time	milliseconds	fastestLap	fastestLapRank	fastestLapTime	faste
26280	1	1	26.0	57	1:31:44.742	5504742	39	1	1:32.608	
26281	5	2	18.0	57	+22.457	5527199	40	4	1:34.364	
26282	4	3	15.0	57	+25.110	5529852	44	6	1:34.507	
26283	2	4	12.0	57	+39.669	5544411	36	2	1:34.090	
26284	3	5	10.0	57	+46.788	5551530	40	12	1:35.065	

```
# Replace '\\N' with np.nan in the respective columns
df['time'] = df['time'].replace("\\N", np.nan)
df['milliseconds'] = df['milliseconds'].replace("\\N", np.nan)
df['fastestLapRank'] = df['fastestLapRank'].replace("\\N", np.nan)
df['fastestLapTime'] = df['fastestLapTime'].replace("\\N", np.nan)
df['fastestLapSpeed'] = df['fastestLapSpeed'].replace("\\N", np.nan)
df['fastestLap'] = df['fastestLap'].replace("\\N", np.nan)
```

```
df.dropna(inplace=True)
```

```
df.isna().sum()
```

```
grid                        0
positionOrder               0
points                      0
laps                        0
time                        0
milliseconds                0
fastestLap                  0
fastestLapRank              0
fastestLapTime              0
fastestLapSpeed             0
year                        0
gp_name                     0
round                       0
driver                      0
drivers_nationality         0
constructors_name           0
constructors_nationality    0
dtype: int64
```

Our dataset mostly contains objects in the next step we are going to convert some of the columns to float

```
# changing our dataset to float for further processing
df.fastestLapSpeed = df.fastestLapSpeed.astype(float)
df.fastestLapRank = df.fastestLapRank.astype(float)
df.milliseconds = df.milliseconds.astype(float)
```

```
# Resetting the index to ensure clean, sequential indexing
df.reset_index(drop=True, inplace=True)
```

5 Exploratory Data Analysis

5.1 Univariate Analysis

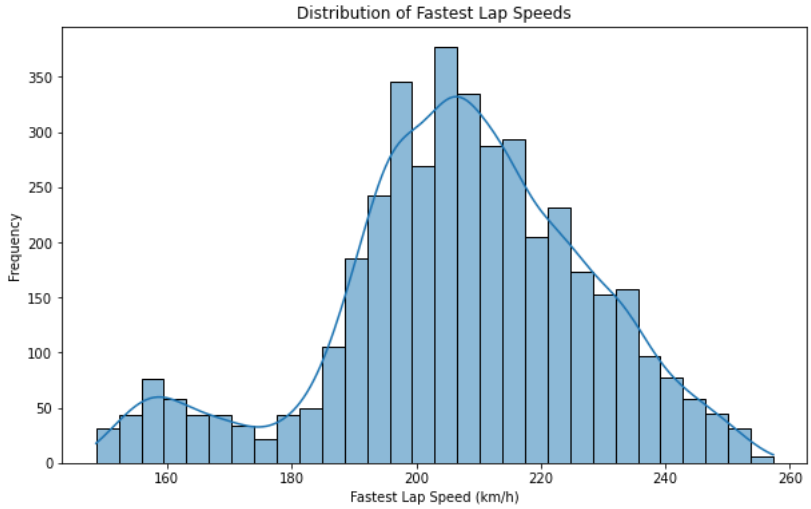
This involves the examination of a single variable. The primary objective is to describe the data and find patterns within it. It helps us understand the central tendency, the distribution and spread of our data.

To do so, we will use summary statistics and distribution plots such as histograms, Box plots and bar plots.

What movies have been the most successfull financially? To answer this question, we look at the box office gross earnings and profits of various movies.

what does the distribtion of the fastest lap speed look like

```
# Distribution of Fastest Lap Speeds
plt.figure(figsize=(10, 6))
sns.histplot(df['fastestLapSpeed'], kde=True, bins=30)
plt.title('Distribution of Fastest Lap Speeds')
plt.xlabel('Fastest Lap Speed (km/h)')
plt.ylabel('Frequency')
plt.show()
```



Summary of Lap Speed Distribution

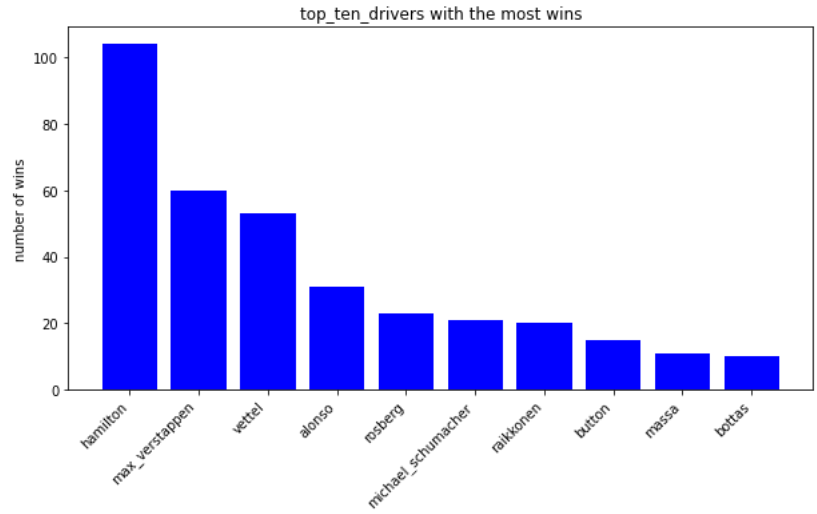
The histogram above displays the distribution of fastest lap speeds, with the x-axis representing speed in km/h and the y-axis showing the frequency of occurrences. The distribution peaks around 200 km/h, indicating that this is the most common fastest lap speed in the dataset. The data is slightly left-skewed, with a broader spread ranging from about 75 km/h to 250 km/h. The overlaid kernel density estimate (KDE) curve helps to smooth out the histogram, providing a clearer view of the underlying distribution. Most lap speeds fall between 175 km/h and 225 km/h, with a notable concentration around 200 km/h.

```
# Gp winners drives
driver_winner = (df.loc[df["positionOrder"] == 1]
                 .groupby("driver")
                 .size()
                 .reset_index(name='wins')
                 .sort_values(by='wins', ascending=False))
```

Who are the top 10 gp winners of all times?

```
# visualizing drivers with the most wins
top_ten_drivers = driver_winner.head(10)
plt.figure(figsize=(10,5))
plt.bar(top_ten_drivers["driver"],top_ten_drivers["wins"],color = "blue")
plt.xticks(rotation=45,ha="right")
plt.ylabel("number of wins")
plt.title("top_ten_drivers with the most wins")
```

```
Text(0.5, 1.0, 'top_ten_drivers with the most wins')
```



From the above visualization:

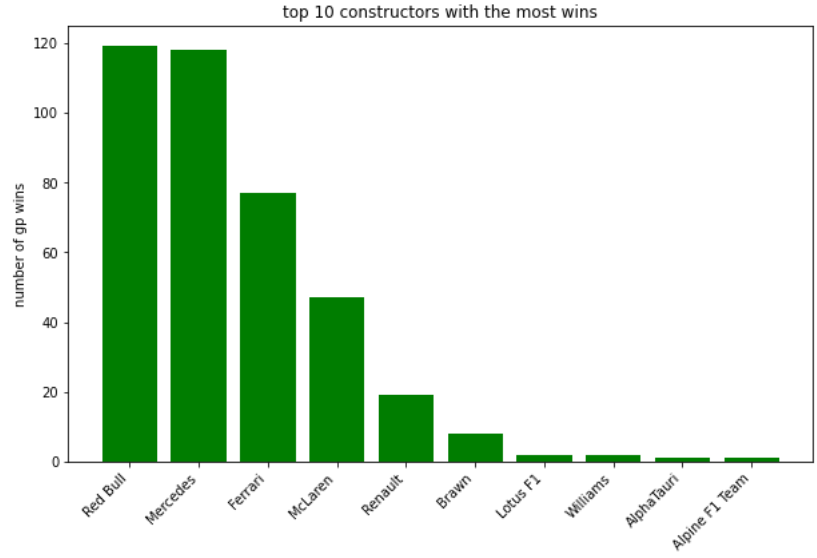
- Hamilton has the most wins, with a bar reaching slightly 104 wins.
- Michael Schumacher is the second, with around 91 wins.
- Max Verstappen is third, with around 61 wins.
- Vettel and Prost follow, each with around 50-60 wins.
- Senna and Alonso are next, with around 40-50 wins.
- Mansell, Stewart, and Lauda are at the bottom of the top ten, each with around 30-40 wins.

who are the top 10 constructors with the most wins ?

```
# GP constructors winners
constructors_winners = (df.loc[df["positionOrder"] == 1]
                .groupby("constructors_name")
                .size()
                .reset_index(name='wins')
                .sort_values(by='wins', ascending=False))
```

```
# Visualizing constructors with the most wins
top_ten_constructors_winners = constructors_winners.head(10)
plt.figure(figsize=(10,6))
plt.bar(top_ten_constructors_winners["constructors_name"],top_ten_constructors_winners["wins"],color = "green")
plt.xticks(rotation=45,ha="right")
plt.ylabel("number of gp wins")
plt.title("top 10 constructors with the most wins")
```

```
Text(0.5, 1.0, 'top 10 constructors with the most wins')
```



From the above visualization

- ferrari has the most wins with approximately 246 wins

- MCLaren is the second in place with with approximately 180 wins
- Mercedes comes in the third place with 127 wins
- red bull comes in the forth place with 120 wins
- williams and Team Lotus are the next with 30 - 50 wins
- Benetton,Brabham,Tyrrel are in the buttom with 20 - 27 wins

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4121 entries, 0 to 4120
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   grid                     4121 non-null   int64  
 1   positionOrder            4121 non-null   int64  
 2   points                   4121 non-null   float64
 3   laps                     4121 non-null   int64  
 4   time                     4121 non-null   object 
 5   milliseconds             4121 non-null   float64
 6   fastestLap               4121 non-null   object 
 7   fastestLapRank           4121 non-null   float64
 8   fastestLapTime           4121 non-null   object 
 9   fastestLapSpeed          4121 non-null   float64
 10  year                     4121 non-null   int64  
 11  gp_name                  4121 non-null   object 
 12  round                    4121 non-null   int64  
 13  driver                   4121 non-null   object 
 14  drivers_nationality      4121 non-null   object 
 15  constructors_name        4121 non-null   object 
 16  constructors_nationality  4121 non-null   object 
dtypes: float64(4), int64(5), object(8)
memory usage: 547.4+ KB
```

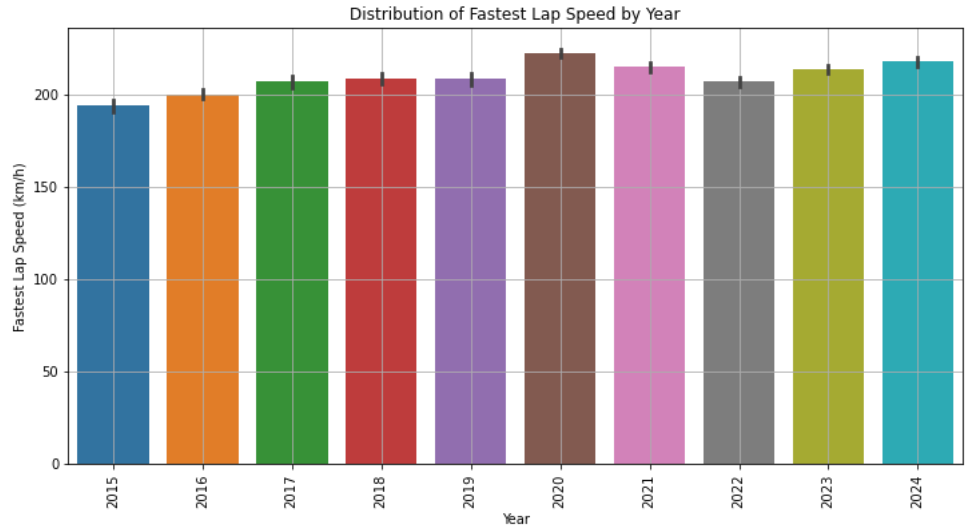
Bivariate analyis

Bivariate analysis is a statistical method of two to variables to understand the relationship between the two variables

```
# checking for the corr between different various numeric columns
corr_matrix = abs(df.corr())
corr_matrix
```

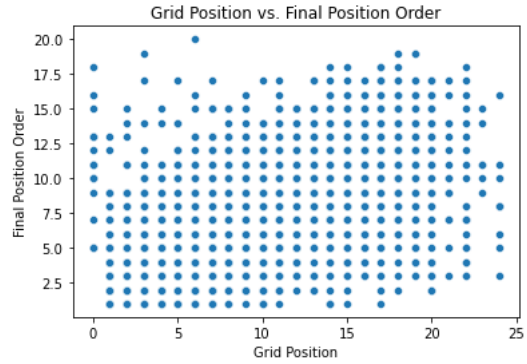
	grid	positionOrder	points	laps	milliseconds	fastestLapRank	fastestLapSpeed	year	
grid	1.000000	0.669152	0.596262	0.095706	0.091377	0.516157	0.036177	0.035810	0.0
positionOrder	0.669152	1.000000	0.825528	0.121220	0.081363	0.654678	0.031641	0.113689	0.0
points	0.596262	0.825528	1.000000	0.069892	0.010395	0.569063	0.009504	0.196484	0.0
laps	0.095706	0.121220	0.069892	1.000000	0.144447	0.091732	0.450862	0.053442	0.0
milliseconds	0.091377	0.081363	0.010395	0.144447	1.000000	0.070795	0.483307	0.093636	0.0
fastestLapRank	0.516157	0.654678	0.569063	0.091732	0.070795	1.000000	0.088942	0.095568	0.0
fastestLapSpeed	0.036177	0.031641	0.009504	0.450862	0.483307	0.088942	1.000000	0.100585	0.0
year	0.035810	0.113689	0.196484	0.053442	0.093636	0.095568	0.100585	1.000000	0.0
round	0.034205	0.033863	0.005076	0.040843	0.044704	0.024445	0.031606	0.044434	1.0

```
plt.figure(figsize=(12, 6))
last_ten_years = df[df["year"]>2014]
sns.barplot(x='year', y='fastestLapSpeed', data=last_ten_years)
plt.title('Distribution of Fastest Lap Speed by Year')
plt.xlabel('Year')
plt.ylabel('Fastest Lap Speed (km/h)')
plt.xticks(rotation=90)
plt.grid(True)
plt.show()
```



What is the impact of grid position to the winning position

```
# plotting a  Scatter plot
sns.scatterplot(x='grid', y='positionOrder', data=df)
plt.title('Grid Position vs. Final Position Order')
plt.xlabel('Grid Position')
plt.ylabel('Final Position Order')
plt.show()
```



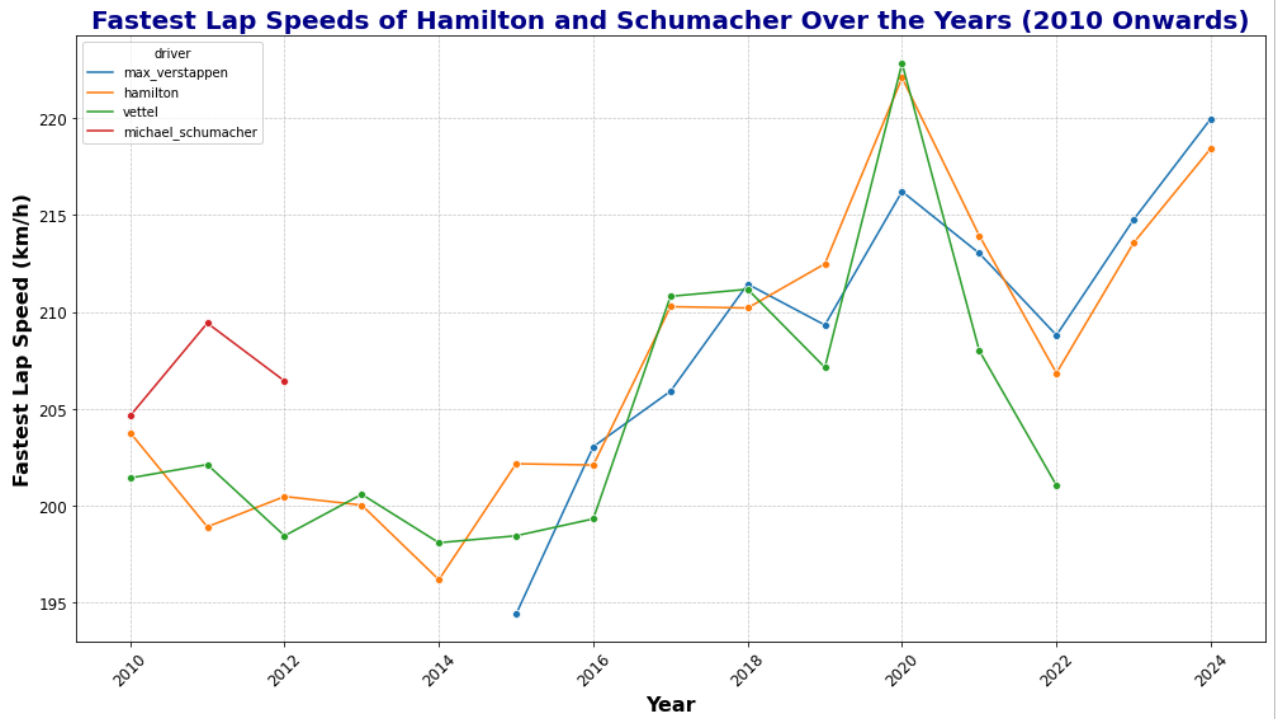
The scatter plot illustrates that while starting grid position has an influence on the final finishing order, it is not the sole determinant. There is considerable variability, meaning that a driver starting in a poor grid position can still achieve a strong finish, and vice versa. This plot underscores the unpredictable and competitive nature of racing, where multiple factors influence the final outcome.

Multivariate Analysis

Multivariate analysis involves examining more than two variables simultaneously to understand the relationships between them.

```
import seaborn as sns
import matplotlib.pyplot as plt

# Filter data from 2010 onwards
df_2010_onwards = df[df['year'] >= 2010]

# Filter the dataframe to include only Lewis Hamilton's and Michael Schumacher's data
df_filtered = df_2010_onwards[df_2010_onwards['driver'].isin(['hamilton', 'michael_schumacher','max_verstappen',

# Plotting
plt.figure(figsize=(14, 8))
sns.lineplot(x='year', y='fastestLapSpeed',hue='driver',data=df_filtered,marker='o',ci=None)

# Add titles and labels with enhanced formatting
plt.title('Fastest Lap Speeds of Hamilton and Schumacher Over the Years (2010 Onwards)', fontsize=20, fontweight
```

```
plt.xlabel('Year', fontsize=16, fontweight='bold')
plt.ylabel('Fastest Lap Speed (km/h)', fontsize=16, fontweight='bold')

# Enhance the axis ticks
plt.xticks(fontsize=12, rotation=45)
plt.yticks(fontsize=12)

# Add gridlines for better readability
plt.grid(True, linestyle='--', linewidth=0.7, alpha=0.7)

# Show the plot
plt.tight_layout()  # Adjusts plot to ensure everything fits without overlap
plt.show()
```



The above graph shows fastest lap time

- Hamilton has shown remarkable consistency and growth, particularly from 2015 onwards, reflecting his dominance in Formula 1 during this period.
- Schumacher's graph reflects the winding down of his career, with stable but non-competitive speeds after 2012.
- Vettel peaked around 2019 but has seen a decline in the following years, suggesting challenges in maintaining peak performance.
- Verstappen showed strong performance early on but has experienced a significant drop after 2019, which could be due to various factors including car performance, team strategies, or external conditions.

Modelling

Logistic Regression

data preparation

We are going to use classification modelling so we are going to convert our target variable to categories

```
def categorize_position(position):
    if position <= 3:
        return 'Top 3'
    elif position <= 10:
        return 'Top 10'
    else:
        return 'Outside Top 10'
```

```
df['position_category'] = df['positionOrder'].apply(categorize_position)

df.head()
```

	grid	positionOrder	points	laps	time	milliseconds	fastestLap	fastestLapRank	fastestLapTime	fastestLa
0	1	1	26.0	57	1:31:44.742	5504742.0	39	1.0	1:32.608	
1	5	2	18.0	57	+22.457	5527199.0	40	4.0	1:34.364	
2	4	3	15.0	57	+25.110	5529852.0	44	6.0	1:34.507	
3	2	4	12.0	57	+39.669	5544411.0	36	2.0	1:34.090	
4	3	5	10.0	57	+46.788	5551530.0	40	12.0	1:35.065	

next step is to split our dataset into x and y

```
# splitting the dataset into x(features) and y(target)
X = df.drop(columns=["position_category","positionOrder"])
y = df["position_category"]
y
```

```
0        Top 3
1        Top 3
2        Top 3
3       Top 10
4       Top 10
         ...
4116    Top 10
4117    Top 10
4118    Top 10
4119    Top 10
4120    Top 10
Name: position_category, Length: 4121, dtype: object
```

Encoding the categorical features for modelling

```
# Encoding categorical features
from sklearn.preprocessing import OneHotEncoder,LabelEncoder
categorical_features = ["gp_name","driver","drivers_nationality","constructors_name","constructors_nationality"]
encoder = OneHotEncoder(sparse=False,drop='first')
encoded_features = encoder.fit_transform(X[categorical_features])
encoded_df = pd.DataFrame(encoded_features, columns=encoder.get_feature_names(categorical_features))

# Combine encoded features with the original DataFrame
X = X.drop(columns=categorical_features).join(encoded_df)
```

```
# encoding the target_variable
y = pd.get_dummies(y,prefix="position").astype(int)
```

```
y
```

	position_Outside Top 10	position_Top 10	position_Top 3
0	0	0	1
1	0	0	1
2	0	0	1
3	0	1	0
4	0	1	0
...	...	...	...
4116	0	1	0
4117	0	1	0
4118	0	1	0
4119	0	1	0
4120	0	1	0

4121 rows × 3 columns

splitting the dataset into Training set and Test set

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=0)
x_train
```

	grid	points	laps	time	milliseconds	fastestLap	fastestLapRank	fastestLapTime	fastestLapSpeed	year
3261	11	0.0	57	+48.822	5800111.0	53	13.0	1:39.704	195.663	2009
1160	10	4.0	78	+55.200	6263637.0	43	3.0	1:15.607	158.890	2019
399	3	10.0	71	+40.845	7049739.0	67	4.0	1:14.007	209.607	2023
1411	15	4.0	67	+31.750	5581595.0	64	17.0	1:17.941	211.267	2018
2481	22	4.0	58	+39.458	5689023.0	46	12.0	1:30.843	210.151	2012
...	...	...	...	...	...	...	...	...	...	...
1033	16	2.0	60	+39.081	5788722.0	54	14.0	1:31.562	202.407	2020
3264	13	0.0	57	+1:26.519	5837808.0	55	1.0	1:38.683	197.687	2009
1653	6	10.0	55	+46.269	5700331.0	43	5.0	1:42.028	195.969	2017
2607	11	0.0	67	+76.829	5542691.0	44	18.0	1:20.066	205.660	2012
2732	6	4.0	56	+73.792	5829061.0	53	16.0	1:41.048	196.409	2012

3296 rows × 207 columns

Feature scalling

```
from sklearn.preprocessing import StandardScaler
numeric_columns = x_train.select_dtypes(include=['number']).columns
x_train_numeric = x_train[numeric_columns]
x_test_numeric = x_test[numeric_columns]
sc = StandardScaler()
x_train_scaled = sc.fit_transform(x_train_numeric)
x_test_scaled = sc.transform(x_test_numeric)
x_train_scaled_df = pd.DataFrame(x_train_scaled, columns=numeric_columns, index=x_train.index)
x_test_scaled_df = pd.DataFrame(x_test_scaled, columns=numeric_columns, index=x_test.index)
```

Training the model

```
from sklearn.linear_model import LinearRegression
linear_model = LinearRegression()
```

```
linear_model.fit(x_train_scaled,y_train)
```

```
LinearRegression()
```

predicting a new result

```
y_pred = linear_model.predict(x_test_scaled)
y_pred
```

```
array([[ 0.32582757,  0.81076172, -0.16173196],
       [ 0.23451898,  0.63864258,  0.06348777],
       [ 0.39662835,  0.70700195, -0.0859263 ],
       ...,
       [ 0.44478509,  0.84958008, -0.27434182],
       [ 0.10787103,  0.67868164,  0.22028708],
       [ 0.33638665,  0.76461914, -0.04729104]])
```

```
y_test
```

	position_Outside Top 10	position_Top 10	position_Top 3
3237	1	0	0
45	0	1	0
2776	1	0	0
2954	0	1	0
3681	1	0	0
...	...	...	...
1944	0	1	0
3420	0	1	0
1351	1	0	0
1422	0	1	0
2678	1	0	0

825 rows × 3 columns

```
from sklearn.metrics import confusion_matrix,accuracy_score
```

```
import numpy as np
y_test_single = y_test.values.ravel()
y_pred_single = (y_pred >= 0.5).astype(int).ravel()
cm = confusion_matrix(y_test_single, y_pred_single)
```

```
from sklearn.metrics import classification_report

report = classification_report(y_test_single, y_pred_single)
print(report)
```

```
              precision    recall  f1-score   support

           0       0.87      0.88      0.87      1650
           1       0.76      0.73      0.74       825

    accuracy                           0.83      2475
   macro avg       0.81      0.81      0.81      2475
weighted avg       0.83      0.83      0.83      2475
```

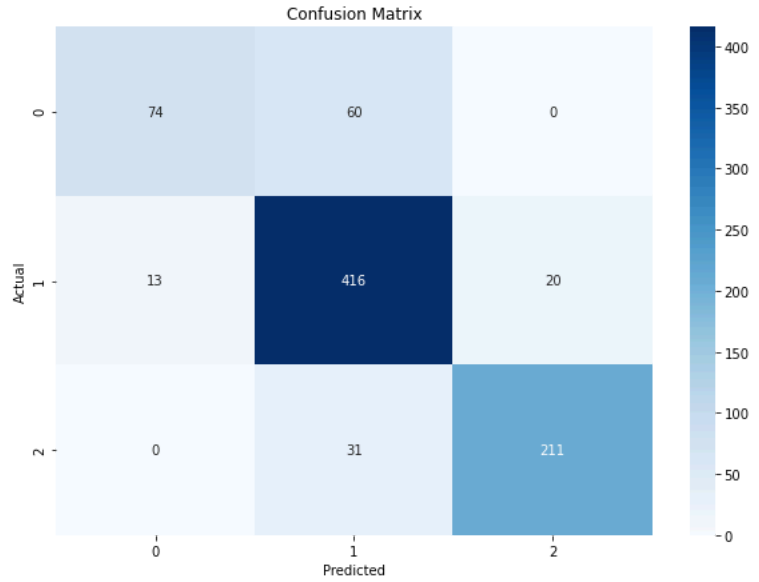
```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Convert y_train and y_test from one-hot encoded back to single labels
y_train_single = np.argmax(y_train.values, axis=1)
y_test_single = np.argmax(y_test.values, axis=1)

# Initialize Logistic Regression model
logreg = LogisticRegression(max_iter=1000, solver='liblinear')

# Fit the model on the training data
logreg.fit(x_train_scaled_df, y_train_single)
```

```
# Predict on the test data
y_pred = logreg.predict(x_test_scaled_df)
```

```
# Calculate accuracy
accuracy = accuracy_score(y_test_single, y_pred)
print(f'Accuracy: {accuracy:.4f}')
```

```
Accuracy: 0.8497
```

```
# Confusion Matrix
cm = confusion_matrix(y_test_single, y_pred)
plt.figure(figsize=(10,7))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=np.unique(y_test_single), yticklabels=np.unique(y
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
print("Confusion Matrix:")
print(cm)

print("\nClassification Report:")
cr = classification_report(y_test_single, y_pred)
print(cr)
```




```
Confusion Matrix:
[[ 74  60   0]
 [ 13 416  20]
 [  0  31 211]]

Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.55      0.67       134
           1       0.82      0.93      0.87       449
           2       0.91      0.87      0.89       242

    accuracy                           0.85       825
   macro avg       0.86      0.78      0.81       825
weighted avg       0.85      0.85      0.84       825
```

Accuracy: 85% Class 0: Precision 85%, Recall 55%, F1-Score 67% Class 1: Precision 82%, Recall 93%, F1-Score 87% Class 2: Precision 91%, Recall 87%, F1-Score 89% Key Insight: The model performs best on Class 2, with high precision and recall. Class 0 shows lower recall, indicating room for improvement.

```
# Classification Report
report = classification_report(y_test_single, y_pred, target_names=[str(i) for i in np.unique(y_test_single)])
print(report)
```

```
              precision    recall  f1-score   support

           0       0.85      0.55      0.67       134
```

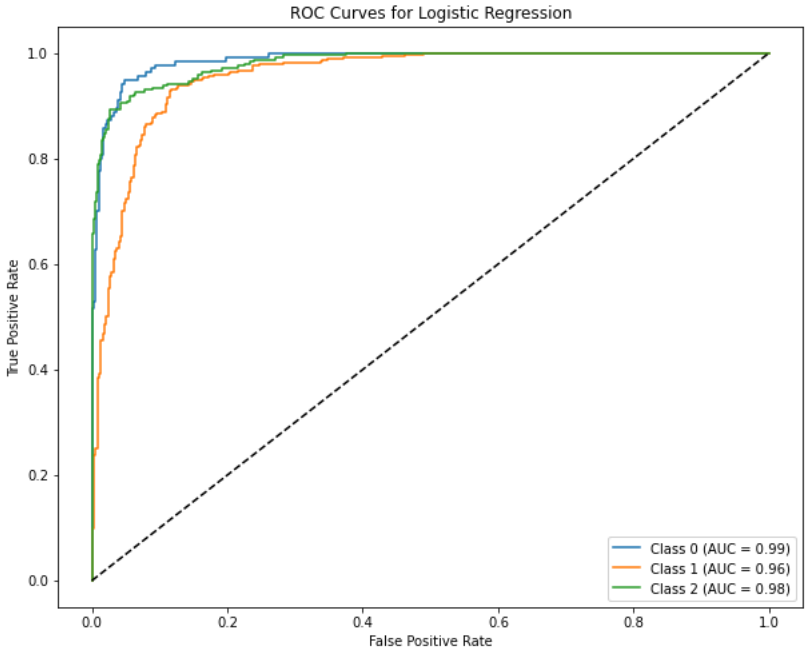
```
              1       0.82      0.93      0.87       449
              2       0.91      0.87      0.89       242

       accuracy                           0.85       825
      macro avg       0.86      0.78      0.81       825
   weighted avg       0.85      0.85      0.84       825
```

Accuracy: 85%

- Class 0: Precision 85%, Recall 55%, F1-Score 67%
- Class 1: Precision 82%, Recall 93%, F1-Score 87%
- Class 2: Precision 91%, Recall 87%, F1-Score 89%
- Key Insight: The model performs best on Class 2, with high precision and recall. Class 0 shows lower recall, indicating room for improvement.

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt

# Step 1: Initialize Logistic Regression model
logistic_model = LogisticRegression()

# Step 2: Fit the model (Assuming you've already split and scaled your data)
logistic_model.fit(x_train_scaled_df, y_train_single)

# Step 3: Predict probabilities for the test set
y_test_proba = logistic_model.predict_proba(x_test_scaled_df)


fpr = {}
tpr = {}
roc_auc = {}

from sklearn.metrics import roc_curve, auc

fpr = {}
tpr = {}
roc_auc = {}

for i in range(len(logistic_model.classes_)):
    # Calculate ROC curve for each class
    fpr[i], tpr[i], _ = roc_curve(y_test_single == i, y_test_proba[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])

# Plotting ROC Curves
plt.figure(figsize=(10, 8))
for i in range(len(logistic_model.classes_)):
    plt.plot(fpr[i], tpr[i], label=f"Class {i} (AUC = {roc_auc[i]:.2f})")
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves for Logistic Regression")
plt.legend(loc="lower right")
plt.show()
```

```
/opt/anaconda3/envs/learn-env/lib/python3.8/site-packages/sklearn/linear_model/_logistic.py:762: ConvergenceWarni
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```



The ROC curves suggest that the logistic regression model is effective in distinguishing between the different classes, with particularly strong performance for Classes 0 and 2. The model may be slightly less effective for Class 1, but it still performs well. These curves can help in selecting the optimal threshold for classification or in comparing different models' performance on the same dataset.

Hyperparameter tuning

```
#Hyperparameter Tuning

from sklearn.model_selection import GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report


model = RandomForestClassifier(random_state=42)

param_grid = {
    'n_estimators': [100, 200, 300],
    'max_depth': [None, 10, 20, 30],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'bootstrap': [True, False]
}

grid_search = GridSearchCV(estimator=model, param_grid=param_grid, cv=5, n_jobs=-1, verbose=2, scoring='accuracy
grid_search.fit(x_train_scaled, y_train)

print("Best Hyperparameters:", grid_search.best_params_)

y_pred = grid_search.best_estimator_.predict(x_test_scaled)
print(classification_report(y_test, y_pred))
```

```
Fitting 5 folds for each of 216 candidates, totalling 1080 fits
[Parallel(n_jobs=-1)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=-1)]: Done  25 tasks      | elapsed:   20.8s
[Parallel(n_jobs=-1)]: Done 146 tasks      | elapsed:  1.1min
[Parallel(n_jobs=-1)]: Done 349 tasks      | elapsed:  2.1min
[Parallel(n_jobs=-1)]: Done 632 tasks      | elapsed:  3.9min
[Parallel(n_jobs=-1)]: Done 997 tasks      | elapsed:  6.4min
[Parallel(n_jobs=-1)]: Done 1080 out of 1080 | elapsed:  7.0min finished
Best Hyperparameters: {'bootstrap': False, 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_e
              precision    recall  f1-score   support

           0       0.95      0.88      0.91       134
```

```
              1       0.93      0.98      0.95       449
              2       1.00      0.91      0.95       242

      micro avg       0.95      0.94      0.95       825
      macro avg       0.96      0.92      0.94       825
   weighted avg       0.95      0.94      0.95       825
    samples avg       0.94      0.94      0.94       825

/opt/anaconda3/envs/learn-env/lib/python3.8/site-packages/sklearn/metrics/_classification.py:1221: UndefinedMetri
  _warn_prf(average, modifier, msg_start, len(result))
```

The model after hyperparameter tuning is performing exceptionally well, with high precision, recall, and F1-scores across all classes. The macro and weighted averages being close suggests consistent performance across different classes. A 94% to 96% performance range in these metrics indicates that your model is likely very effective in predicting the categories you've set (Top 1, Top 3, Out of Top 10).

Decision Tree Regression

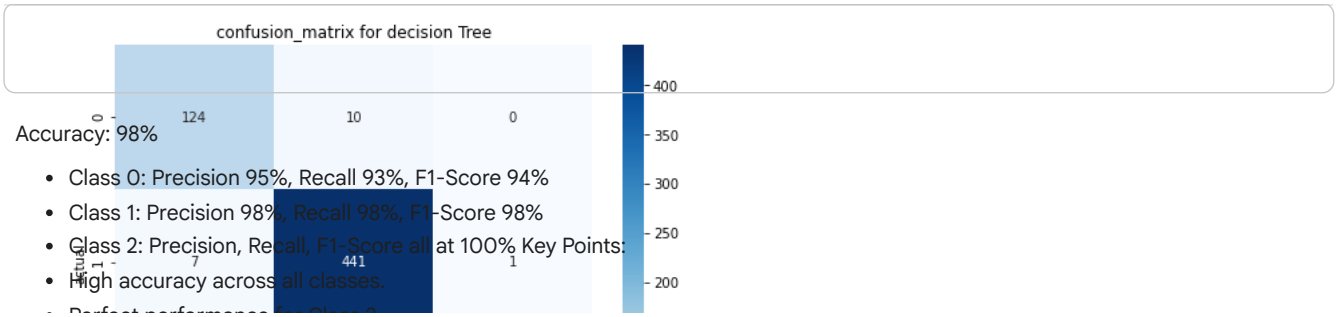
```
# importing necessary libraries
from sklearn.tree import DecisionTreeClassifier,plot_tree
from sklearn.metrics import classification_report,confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Training the decision tree classifier
decision_tree = DecisionTreeClassifier(random_state=0)
decision_tree.fit(x_train_scaled,y_train)

y_pred_dt = decision_tree.predict(x_test_scaled)
y_pred_single = np.argmax(y_pred_dt,axis=1)
```

```
# confusion_matrix
cm_dt = confusion_matrix(y_test_single,y_pred_single)
cm_dt
```

```
array([[124,  10,   0],
       [  7, 441,   1],
       [  0,   0, 242]])
```

```
# plotting the confusion matrix
plt.figure(figsize=(8,6))
sns.heatmap(cm_dt,annot=True,fmt="d",cmap="Blues")
plt.title("confusion_matrix for decision Tree")
plt.xlabel("Predicted")
plt.ylabel("actual")
plt.show()

print(classification_report(y_test_single,y_pred_single))
```



Accuracy: 98%

- Class 0: Precision 95%, Recall 93%, F1-Score 94%
- Class 1: Precision 98%, Recall 98%, F1-Score 98%
- Class 2: Precision, Recall, F1-Score all at 100% Key Points:
- High accuracy across all classes.
- Perfect performance for Class 2